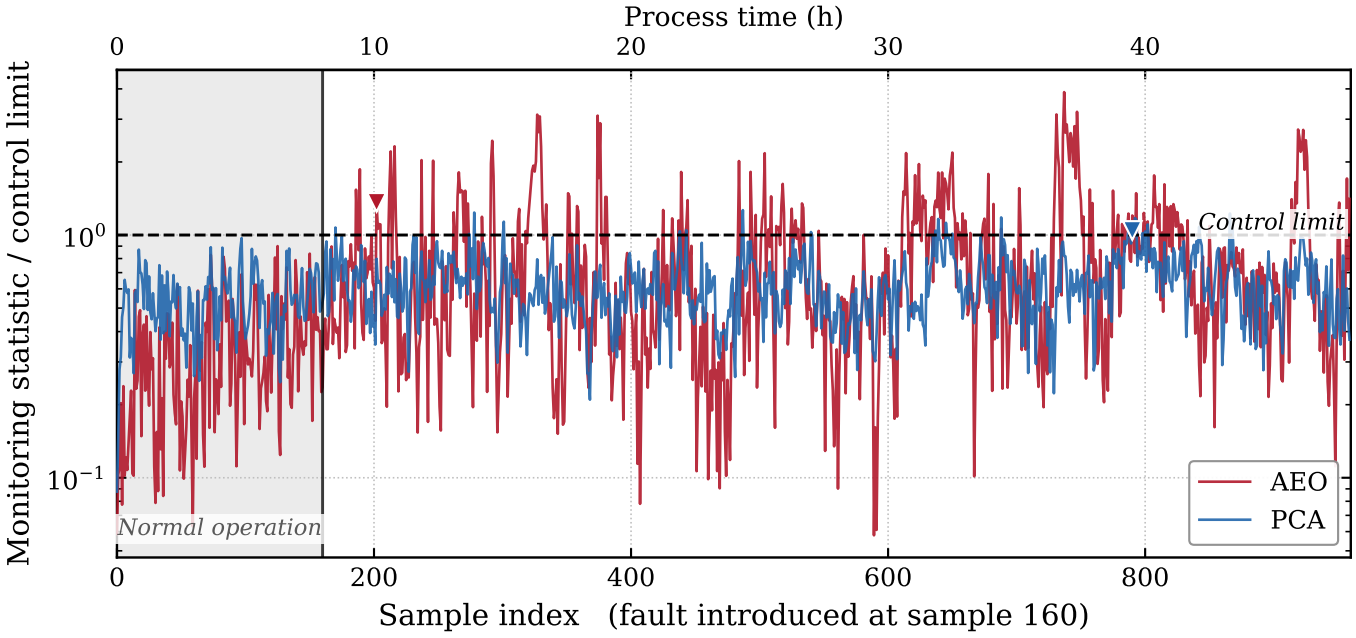
Fault Detection Performance for IDV(16)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	81.9 samples (246 min)	MDR 0.776	detected in 7/7 runs
PCA	delay	246.3 samples (739 min)	MDR 0.952	detected in 6/7 runs

AEO detects 164.5 samples (493 min) earlier; AEO missed-detection rate lower by 19%